Project

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Project Information

MMA 609 Winter 2026

In this document:

Important dates

Friday March 6 11:59 pm	Initial selection of topics (Google form linked below)
Monday March 9 11:59 pm	Final assignment of topics complete
Monday March 16 11:59 pm	Pitch presentation due (on Canvas)
Wednesday April 8 11:59 pm	Presentation slides due (on Canvas)
Thursday April 9	Project presentations in class

components, equally weighted:

1. Pitch presentation: Presentation to the class about their intended project.
2. Progress presentation: A little past the midpoint for the project, an update to the class around the progress thus far and the plans for completion.
3. Project: Students will have to submit a final notebook (or other code repository) showing their work.
4. Final Report: Students will present a final report reflecting on the final project.

More detail on these components are given in later sections.

Project topics

See the pdf file on Canvas (in the Project module) for a list of potential topics. Select your topic by Friday March 6th 11:59 pm using this form: https://forms.gle/oi4UYNzkYxrJ2NQH9 . When you select the topic, you will also select the Responsible AI perspective you will use for your analysis: privacy, fairness/bias, or explainability/transparency. This new form only shows you the topics that remain after other students have made their selection. If you don't make an initial selection by the date below, I will select for you.

Important dates:

Friday March 6 11:59pm	Initial selection of topics
Monday March 9 11:59pm	Final assignment of topics complete

Responsible AI aspect you selected. The project consists of a) a conceptual analysis of the scenario with the perspective chosen, b) implementation on a python notebook with a selected dataset, and c) a report that combines the two and provides a conclusion from the point of view of the persona you select.

Stakeholder

For the analysis (non-technical) part of the project you will assume a stakeholder perspective. This persona will be used only in your final conclusion and the final *takeaway* point you will make. The goal is to exercise executive analysis and decision-making that you might see in a practical and real-life management scenario. For example as a data scientist, you might make technical conclusions on the feasibility of a given privacy preserving technique with respect to the company's goals and capacity, or as the Chief Risk Officer you might make conclusions on company guidelines and policy that should be set based on your analysis. Example stakeholders or personas to assume:

- Data scientist
- Chief Risk Officer
- Chief Compliance Officer
- Privacy and Data Officer
- Chief Technology Officer

Implementation of your project, and the final report and presentation should be done to support your chosen stakeholder.

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1. Pitch Presentation

Deadline: Monday March 16 11:59 pm.
Submit on Canvas:

- A small report (250 words) describing your project. (pdf)
- A short recorded video (2 minutes max) where you present your project. (mp4)
- Your slide deck. (pdf)

Pitch report. This is a very small description, maximum 250 words, that describes your project: the scenario you selected, the aspect of Responsible AI you will use for the study, what datasets and models you will use, and what you hope to achieve overall.
Pitch presentation. You will record a presentation of maximum 2 minutes length, where you present your project, with slides. You should present on what is in the pitch report description above, and provide some motivation for the project (why is this important?).

2. Final presentation

During the class on **April 9th**, each student will present their final project. The presentation will be a maximum of 3:30 minutes long. I will publish the random order of presentations the week of the presentations.
Slides should be submitted on Canvas by 11:59 pm on April 8th 11:59pm. You can use Powerpoint, Keynote, Google slides or other online presentation methods, however the final set must be converted to pdf. The file you submit must have the following naming convention:

The presentations will be in a 3:30-minute executive lightning talk format.

Presentation guidelines

1. The "4-Slide / 4-Minute" Rule

In a 3:30-minute window, you cannot cover every line of code. You must distill your project into four key slides, spending roughly 50 seconds on each, and likely more time on slides 2 and 3 than the rest:

- **Slide 1: The business case & persona.** State the scenario and your specific RAI lens (Privacy, Fairness, or Explainability). Introduce your **Stakeholder Persona** immediately so the audience knows whose perspective you are speaking from.
- **Slide 2: The technical approach.** Show a snippet of your logic or a flow diagram. Do not show raw code; show the *method* (e.g., "We applied an ϵ-budget of 1.0 using the Laplace mechanism"). Talk about the dataset(s) you used and any changes or additions you made. Talk about why this is a suitable dataset and method.
- **Slide 3: Key visual evidence.** Present your primary RAI metric(s). This should be a "Management-Ready" chart (e.g., a SHAP waterfall plot, a Fairness Disparate Impact chart, or a Privacy-Utility curve).
- **Slide 4: The stakeholder recommendation.** This is your "Executive Takeaway." Based

- **Minimize text on slides:** Use a minimum **24pt font**. If you have to read the slide to us, you are losing time and the audience is losing interest. Use images and diagrams to convey your message as much as possible. On your charts and plots ensure axes are labeled and readable from the back of the room.
- **One insight per slide:** Each slide should have a clear headline that states the question or conclusion (e.g., *"Model shows 20% bias against Group X"* instead of *"Fairness Results"*).

3. The Persona

You are not presenting to the professor; you are presenting to your chosen **stakeholder**.

- **Tone:** Adopt the expected professional tone. A Data Scientist might focus on "feasibility," while a Chief Compliance Officer focuses on "liability."
- **The final word:** Your very last sentence must be a direct recommendation to the board/management.

Classroom Logistics & Rules

To get through everyone's presentations, the following **hard rules** will be in place:

- **The "On-Deck" system:** All slides will be projected from the desktop computer, with slides in the shared folder. Slide decks will be in the folder in order. Everyone should be ready: while Student A is silent "30-second warning." I will wave a sheet with the number 30 on it. At 3:30 minutes, the presentation will end even if you are mid-sentence. The next student will come up, close your presentation and start theirs. The next student's clock starts 10 seconds after the end of the previous student's 3:30 minutes. Your presentation can be only 3 minutes long, but not longer than 3:30 minutes.

- ~~**One question:** There will be 20 seconds for one question while the next student gets ready.~~ There will be no time for questions.
- **Single file submission:** All slide decks (PDF or PowerPoint) must be submitted by 11:59 pm the previous night on Canvas. We will use the classroom desktop for all presentations. To avoid surprise technical difficulties, you can check that the slides appear as you intend on the desktop computer before submitting.
- **No Live Demos:** Do not attempt to run your Python notebook live. Use screenshots or recorded GIFs of your outputs in your slides.

Peer Review

Each student will be assigned 3 peers chosen at random. The reviewers will be announced at the start of each presentation - pay attention so that you don't miss your reviews. For each of the 3 peers, you should provide one "pro" comment (one thing they did very well) and one "grow" comment (one thing they can improve on). This will

3. Project Implementation

Submit a final python notebook showing your work. Submit the notebook file, and an html file where you have executed all cells. If you are not working on notebook files but coding in python, then submit a link to the github repository with clear instructions in your readme file on how to execute the code. The github repository should not be edited after submission.

Implementation guidelines

You are not required to code from scratch any component of the project. You can use existing modules and packages, and the help of AI tools. However, you should:

- describe your process of implementation in your project final report,
- cite the sources you used,
- be able to explain your implementation.

In your implementation, the metrics you measure and the methods and techniques you implement should be in line with the core question of your project, and be in line with the support it provides the stakeholder persona you have chosen. For example, if you are a Chief compliance officer who needs to make decisions of approving a product line, the metrics measured and the techniques implemented should provide support for such a decision.
Expectation: you implement the techniques we covered in class for your respective responsible AI aspect (privacy, fairness, explainability). If you implement a technique we did not cover in class, this is fine, but you should properly cite everything. You

full marks.

4. Final Report

Submit your final report on the project by April 10 11:59 pm on Canvas

Overview

The final report should be a professional document (PDF) that integrates your conceptual analysis, technical findings, and stakeholder recommendations.

- Length: 4-6 pages (2,000–3,000 words, excluding appendices/references).
- Format: Professional report (Standard margins, 11pt font, clear headings).
- Deadline: April 10, 11:59 PM via Canvas.
- Naming Convention: LastName_Topic##_FinalReport.pdf

Report Structure

1. Executive Summary (1 Page)

Provide a high-level overview of the problem, the Responsible AI (RAI) lens applied, the key technical finding, and the final recommendation. This is written for the CEO or Board—assume they will not read the rest of the report.

2. Problem Context & Persona Definition

Scenario Analysis: Describe the business scenario and the specific RAI responsibilities?

3. Technical Methodology

The RAI Lens: Explain the theoretical foundation of your chosen lens (Privacy, Fairness, or Explainability). Define the core metrics (e.g., Explain the logic of ϵ-Differential Privacy or the math behind Equalized Odds).
Dataset Description: Describe the data used. Why was it suitable? What were its limitations? Describe your synthetic data generation process if used.
Implementation Process: Summarize the steps taken in your Python notebook. (Do not paste code here; describe the logic).

4. Results & Business Interpretation

Key Findings: Present the primary charts/tables from your implementation.
Translation: Translate technical results into business reality. E.g. If your SHAP analysis shows "Time on App" is the top predictor for credit, what does that mean for the company's reputation? If your privacy budget ϵ is 1.0, what is the actual risk?

5. Final Stakeholder Recommendation

As your persona, provide a clear, actionable decision: Go/No-Go: Should the model be deployed? Mitigation: What specific changes were made? (e.g. To ensure privacy, or fairness) How should the deployment pipeline be further modified?

Citations: APA or IEEE format for all papers, libraries, and AI tools used.
Appendix: Include supplemental charts or a brief "Process Log" of how you used Generative AI to assist in the coding or writing process.